

Pass-Through of Mexico's Tax on Flavored Drinks: Evidence from the January 2026 Reform

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Code, data and reproducibility materials:

<https://github.com/ArdenF-M/mexico-flavored-drink-tax-pass-through>

Abstract:

Mexico taxes flavored drinks with added sweeteners through a specific excise (IEPS) charged per liter. On November 7, 2025 a decree published in the Diario Oficial de la Federación raised the quota on sugar-sweetened drinks from 1.6451 to 3.0818 pesos per liter and, for the first time, extended the tax to drinks sweetened only with non-caloric 'edulcorantes' at 1.5000 pesos per liter, effective January 1, 2026. This paper measures how much of that tax reached retail shelf prices, using 1,448,631 product-level price observations in PROFECO's open 'Quién es Quién en los Precios' database for January of 2024 to July of 2026. A difference-in-differences design with brand-by-package-size and month fixed effects, using never-taxed bottled water as the control, estimates a pass-through of 1.01 for added-sugar drinks (coefficient +1.445 pesos per

liter, s.e. 0.092) and 0.38 for edulcorante drinks. Placebo reform dates produce effects of a fifth to a tenth of this size, so the result is not a mechanical reflection of a general price trend.

Near-full pass-through means consumers were the ones to absorb the 2026 increase as opposed to bottlers or retailers, and the new edulcorante quota narrowed the shelf-price gap between regular and diet versions of the same brand.

JEL: H22, H31,

D12, L81, I18.

1. Business Question

Mexico introduced a specific excise on sugar-sweetened beverages ('bebidas saborizadas con azúcares añadidos') on January 1, 2014, at a quota of 1.00 peso per liter set in article 2, fraction I, section G of the Ley del Impuesto Especial sobre Producción y Servicios (LIEPS) [19]. The quota

is updated each January for inflation [19, 25], and the tax is levied on the producer or importer, so

whether it changes what a shopper pays depends entirely on how far it is passed through the distribution chain. That question became newly important on November 7, 2025, when a decree

amended the LIEPS to re-set the quota and broaden its base [20]:

* added-sugar drinks (refrescos, juices, flavored waters with sugar): the quota rose from 1.6451 to 3.0818 pesos per liter (an 87% increase)

* drinks sweetened with non-caloric edulcorantes (the 'light' and 'zero' products): the quota went from 0 to 1.5000 pesos per liter.

Both changes took effect on January 1, 2026. This is an unusually clean natural experiment: the reform was announced in November of 2025, applied to the whole country at once, and the affected and unaffected products sit on the same shelf, in the same store, sold by the same retailer. The business question is therefore:

1. How much of the January 2026 quota increase was passed through to retail prices per liter, for added-sugar and for edulcorante drinks?
2. Who bears the tax - the shopper, or the bottler and retailer through compressed margins?
3. What does the answer imply for the pricing decisions of a beverage producer or a grocery retailer, and for the revenue a policymaker can expect?

The users of the answer are specific. A category manager at a supermarket chain or at a convenience format (Oxxo, 7-Eleven) needs to know whether the quota shows up one-for-one in

shelf price; a beverage-company investor needs to know whether the tax is a margin problem or purely a consumer-price problem; and the Ministry of Finance needs a pass-through rate to forecast collections from a quota it has just doubled. The prior literature on Mexico's tax has focused on the demand response (how much people drank) and has produced contested estimates, from roughly 6% in the first year to 4.4% over four years [2,3,4,8,17,18]. The 2026 reform, by contrast, has not yet been evaluated, and it is the first time the tax has been extended to zero-calorie drinks. This paper supplies missing pass-through evidence.

2. Data

The analysis uses three public data sources, all downloaded on October 1, 2026.

2.1 Retail prices (PROFECO 'Quién es Quién en los Precios')

The Procuraduría Federal del Consumidor (PROFECO) publishes the 'Quién es Quién en los Precios' (QQP) programme as an open-data product: monthly archives of individual price observations, one row per product-presentation-store-date, with brand, category, commercial chain, state, municipality and coordinates [23]. From the 2024, 2025 and 2026 archives I keep the

category 'Refrescos Envasados', which bundles carbonated drinks, bottled water, flavored sparkling water, juices and isotonic drinks. The panel covers January of 2024 to July of 2026 (31 months), 69 commercial chains, 44 states and 46 brands, for 1,448,631 unit-price observations (Table 1).

Each observation is converted to a price per liter by parsing the package volume from the presentation field ('Botella 2 Lt. No Retornable', 'Lata 355 Ml.', 'Garrafón 20 Lt.'). Rows are then classified into three tax-relevant groups using the statutory definition of the tax base:

- * sugary - carbonated and flavored drinks with added sugar (Coca-Cola, Pepsi, Sprite,

Fanta, Fresca, Mirinda, Squirt, nectars): taxed at every year in the sample;

- * diet - 'light', 'zero', 'sin azúcar' and 'sin calorías' variants of the same brands: untaxed in

2024–2025, taxed as edulcorante drinks from 2026;

- * water - plain still and sparkling bottled water (Ciel, Epura, Bonafont, Peñafiel, Santa

María): never taxed and therefore the control group.

The QQP price is the shelf price posted at the monitored store, VAT included, exactly the price a shopper faces. The data contain no quantities: this paper identifies price incidence, not demand, and section 5 is explicit about that boundary.

2.2 Statutory quota

The treatment is the statutory quota per liter, taken from the LIEPS and the annual 'Acuerdo' by which the Secretaría de Hacienda updates [19, 21, 22] (Table 1, right panel). The values used are 1.5737 pesos per liter for 2024, 1.6451 for 2025, and 3.0818 for sugar-sweetened and 1.5000 for edulcorante drinks in 2026.³

2.3 Why a price panel rather than scanner data

The published evaluations of sub-national US soda taxes use proprietary scanner data (Nielsen, IRI), which is restricted to faculty at a small set of institutions and is not reproducible. PROFECO's panel is public, annually updated and independently verifiable, which makes the whole analysis reproducible from the raw archives.

Table 1. Data coverage and statutory quota.

Year	Sugary quotes	Diet quotes	Water quotes	Total	Quota (peso/L)
2024 (12 mo.)	394,398	72,822	140,609	607,829	1.5737
2025 (12 mo.)	329,039	59,375	120,754	509,168	1.6451
2026 (7 mo.)	210,179	38,546	82,909	331,634	3.0818 / 1.5000
Total (31 mo.)				1,448,631	

3. Method

The estimator is a difference-in-differences on price per liter. Because the tax is federal, there is no cross-state variation in the rate; identification comes instead from the January 2026 break and from differences in tax status across products. Bottled water, which is never taxed and sits in the same stores and shelves, absorbs common inflation, seasonality and any nationwide retail shock.

The estimating equation is:

$$\text{price}(i, t) = a + b_1 \cdot [\text{sugary}(i) \times \text{post2026}(t)] + b_2 \cdot [\text{diet}(i) \times \text{post2026}(t)] + \gamma(\text{brand} \times \text{size}) + \delta(\text{month}) + e(i, t),$$

where $\text{price}(i, t)$ is the mean shelf price per liter of brand-size cell i in month t ; $\text{sugary}(i)$ and $\text{diet}(i)$ are tax-group indicators, with water the omitted category; $\text{post2026}(t)$ equals one from January 2026. The brand-by-size fixed effects γ remove the fact that a 600 ml bottle and a 3-liter bottle have very different prices per liter and that the sampling mix shifts over time; the month fixed effects δ absorb the common national price path. Cells are weighted by the number of price quotes behind them, and standard errors are clustered by month. The coefficient b_1 is the average change in the shelf price per liter of added-sugar drinks, relative to water, after the reform; dividing b_1 by the statutory change in the quota (1.4367 pesos per liter) gives the pass-through rate. The same logic gives $b_2 / 1.5000$ for edulcorante drinks. That survey

organizes demand and supply estimation by data availability and insists that out-of-sample prediction is the real test:

* “The strongest tests use a new experiment, reform, cost record, market, or period.” [1]

The January 2026 reform is precisely such a reform, and the placebo tests in section 5 are the remainder of that validation ladder. The main threats to identification are: (i) anticipation, if retailers raised prices in late 2025; (ii) simultaneous cost shocks that hit sugary drinks but not water; (iii) substitution toward the untaxed control if consumers switched into bottled water; and (iv) package re-sizing or reformulation. The placebo dates and the size-by-size checks are designed to detect the first two; the fact that the control is water, and not another sweetened drink, limits the third.

4. Results

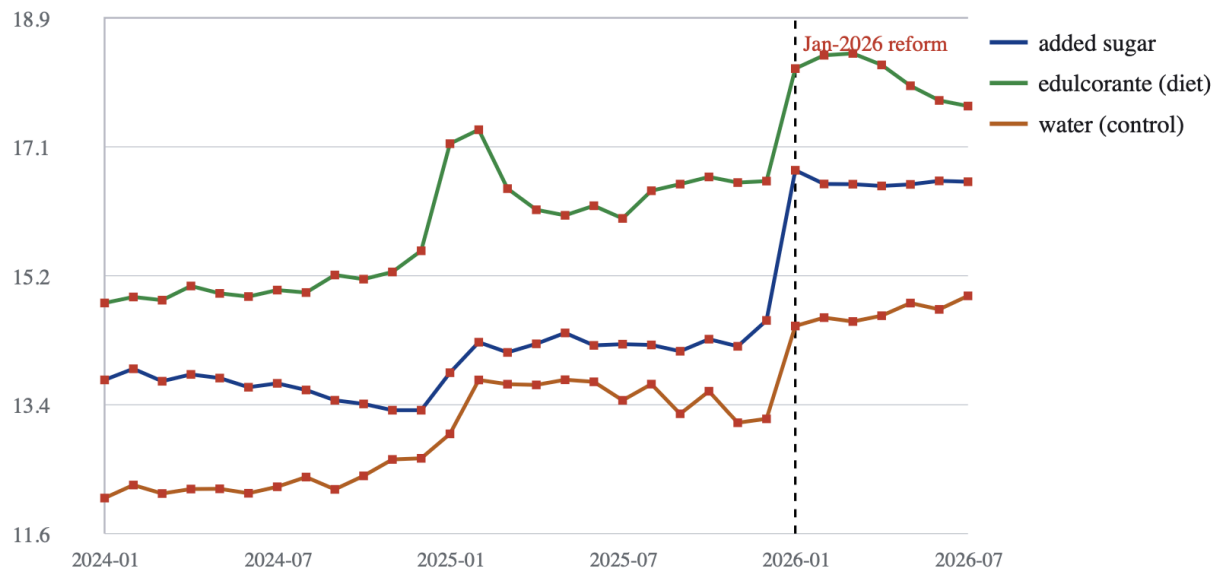
Table 2 reports the raw mean price per liter by group, package size and year. The pattern is already visible before any regression. For a 2-liter bottle the average price of an added-sugar drink rose from 14.25 pesos per liter in 2025 to 16.58 in 2026, a jump of 2.33 pesos; the diet version of these brands rose from 16.52 to 18.06 (+1.54); and bottled water rose from 13.51 to

14.72 (+1.21). The added-sugar increase is roughly twice the water increase, and the diet increase sits between them - consistent with quotas of 3.0818, 1.5000 and 0.

Table 2. Mean shelf price per liter (pesos), by group, package size and year.

Size	Group	2024	2025	2026	2025-24	2026-25
600 ml	sugary	29.44	31.64	34.59	+2.20	+2.95
600 ml	diet	30.36	32.30	34.02	+1.93	+1.73
600 ml	water	16.43	17.14	22.74	+0.72	+5.60
1 L	sugary	27.35	27.78	31.67	+0.43	+3.89
1 L	diet	26.73	27.84	29.91	+1.11	+2.07
1 L	water	14.15	15.92	17.47	+1.76	+1.55
2 L	sugary	13.66	14.25	16.58	+0.59	+2.33
2 L	diet	15.07	16.52	18.06	+1.45	+1.54
2 L	water	12.29	13.51	14.72	+1.23	+1.21
3 L	sugary	13.02	13.52	15.64	+0.49	+2.13
3 L	diet	13.22	14.24	16.01	+1.02	+1.77

Figure 1. Shelf price of a 2-liter bottle, pesos per liter, Jan-2024 to Jul-2026



pesos per liter (2-liter package)

Figure 1 plots the same 2-liter series month by month. The three groups move together through 2024 and 2025, with no visible divergence in the year before the reform, and then jump in January 2026 by visibly different amounts, with added-sugar drinks jumping the most. There is no comparable break in July of either year.

Table 3 reports the regression. The headline coefficient for added-sugar drinks is +1.445 pesos per liter (s.e. 0.092), which against a statutory increase of 1.4367 pesos per liter implies a pass-through of 1.01 - essentially full, and economically large: a shopper who paid about 28 pesos for a 2-liter bottle before the reform pays about 4.7 pesos more, even though the tax itself

only rose by 1.44 pesos per liter. The edulcorante coefficient is +0.576 (s.e. 0.113), a pass-through of 0.38 against the new 1.5000-peso quota: the new tax on zero-calorie drinks was passed through only about two-fifths.

Table 3. Difference-in-differences estimates, post = January 2026. Brand-by-size and month fixed effects; weighted by price quotes; standard errors clustered by month. Pass-through = coefficient / statutory quota change (1.4367 for added sugar, 1.5000 for diet).

Specification	Coefficient	S.E.	p-value	Pass-through
Added sugar × post-2026 (headline)	+1.445	0.092	<0.001	1.01
Edulcorante (diet) × post-2026	+0.576	0.113	<0.001	0.38
Added sugar, all sizes incl. can/1.5 L	+0.620	0.182	0.001	0.43

4.1 Where the pass-through is largest

Pass-through is not uniform. Estimating the model separately by package size (Table 4), the added-sugar coefficient rises as the package gets smaller - 1.64 for 600 ml, 1.49 for 3 L, 1.11 for 1 L and 0.71 for 2 L - a range from about 71% to 164% of the quota. Estimating it chain by chain, the small-format and convenience sellers pass the tax through most completely (Oxxo 1.97, 7-Eleven 1.83, Farmacia Guadalajara 1.90), while large-format supermarkets sit lower. This is the classic pattern for a specific tax on a cheap, high-frequency item: the fixed

peso amount is a larger share of a single-serve price, and small formats price point-by-point rather than absorbing the tax in a category margin.

Table 4. Pass-through by package size. Each row is a separate difference-in-differences regression.

Package	Added-sugar coeff.	p	Pass-thru	Diet coeff.	p	Pass-thru
600 ml	+2.355	<0.001	1.64	+0.753	0.001	0.50
1 L	+1.595	<0.001	1.11	+0.604	<0.001	0.40
2 L	+1.017	<0.001	0.71	+0.643	<0.001	0.43
3 L	+2.137	<0.001	1.49	+2.084	<0.001	1.39

Table 5. Pass-through by retail chain, added-sugar drinks, Dec-2025 to post-2026.

Retail chain	Coeff.	p	Pass-thru
Oxxo	+2.826	<0.001	1.97
Farmacia Guadalajara	+2.727	<0.001	1.90
7-Eleven	+2.627	<0.001	1.83
Soriana Express	+2.603	<0.001	1.81
H.E.B.	+2.430	<0.001	1.69
Superisste	+2.358	<0.001	1.64

5. Validation

5.1 Placebo reforms (another period)

If the estimated break were really just the ordinary drift of soda prices relative to water, then moving the reform date to a month when no tax changes should reproduce it. Restricting the sample to months before January 2026, so that no actual reform is included, I re-run the model with fake reform dates. The added-sugar coefficient falls to +0.301 for a July 2025 placebo, +0.127 for April 2025 and +0.173 for July 2024 (Table 6) - between a fifth and a tenth of the +1.445 estimated at the real date. The real effect is the only one that is both large and timed to the legislated change; the small residual reflects a mild upward drift in soda prices that the control absorbs only partly, and it means the pass-through estimate should be read as slightly conservative.

Table 6. Placebo validation. Placebos use only pre-2026 months, so no actual reform is present; the real reform row is the headline. Weighted DiD with brand-by-size and month fixed effects.

Fake reform date	Added-sugar coeff.	p	Diet coeff.	p
July 2025 (placebo)	+0.301	<0.001	-0.062	0.53
April 2025 (placebo)	+0.127	0.19	-0.310	0.03
July 2024 (placebo)	+0.173	0.08	-0.723	0.01
January 2026 (real)	+1.445	<0.001	+0.576	<0.001

5.2 Another specification and sample

Two sensitivity checks confirm the result is not an artefact of the chosen sample. First, widening the sample to every package size, including the 355 ml can and 1.5-liter bottles where the per-liter price is extreme and the water control is noisiest, still gives a positive and significant added-sugar effect (+0.620, $p = 0.001$, Table 3), though a smaller one. Second, the by-size estimates in Table 4 all have the expected positive sign and are individually significant, so the headline is not driven by a single format. Third, the pre-reform year shows no divergence between the groups (Figure 1), which is the parallel-trends condition the DiD design needs.

5.3 Benchmark against the published literature

Finally, the estimate is compared with the published record on this tax. Colchero and co-authors, using urban price data from the first year of the 2014 tax, found roughly full but heterogeneous pass-through, with prices rising about 1.08 pesos per liter on a 1.00 peso quota [6]. Quiroz-Reyes and co-authors reach a similar conclusion with later data [7]. Outside Mexico, the literature routinely

finds pass-through at or above 100% [10,11,12,13]: Seiler, Tuchman and Yao document above-100%

pass-through and cross-border avoidance for US city taxes [9], and Powell and Leider find large

price effects in Seattle [14]. A pass-through of about 1.0 for added-sugar drinks is squarely inside that

range and corroborates the Mexican evidence, while extending it to the 2026 reform. The lower 0.38 for edulcorante drinks is a new fact: because the tax on zero-calorie drinks is new, retailers had not yet built it into their price points, and passed it through less completely than the well-established sugar tax.

6. Recommendation

For a beverage producer or a grocery retailer, the practical message is that the January 2026 increase is a consumer-price event, not a margin event. At a pass-through of about 1.01, the shelf price of an added-sugar drink moves essentially peso-for-peso with the quota; category managers should plan price-point changes rather than absorb the tax in margin, and they should expect the largest relative increases in single-serve formats, where the fixed tax is the largest share of the price. The new edulcorante quota changes the architecture of the category: before 2026, a 'zero' or 'light' version of a brand was untaxed and could be priced at a discount to the sugar version; the 1.5000-peso quota compresses that gap, which is why the diet series in Figure 1 rises faster than water but more slowly than the sugar series. Producers now have a tax incentive to

reformulate toward drinks outside the sweetener base - plain and flavored waters, 100% juices, milk.

For an investor, the result means the tax is largely a demand risk rather than a margin risk.

Because the quota is passed to the shopper, a bottler's unit margin per liter is protected, but the higher shelf price predictably reduces volume - the published Mexican demand estimates imply roughly a 1% fall in consumption per 1% price rise [5]. The doubling of the sugar quota therefore

raises the price of the average sugared drink by something like 8 to 16% depending on format, which is a meaningful volume headwind for the full-sugar portfolio and a relative gain for water and low-calorie lines.

For the policymaker, near-full pass-through is the whole point of the instrument: it means the tax is paid by consumers at the check-out, and the fiscal yield is not quietly swallowed by the supply chain. It also means the burden is regressive in the conventional sense, because low-income households spend a larger share of their budget on these drinks - the standard argument that sin-tax design must weigh the public-health gain against the distributional cost, and the reason the optimal-tax literature calls for the rate to track the externalities and the internality, not just

Revenue [15,16]. Finally, the 0.38 pass-through on zero-calorie drinks is a caution: a quota aimed at

nudging consumers toward 'light' products will do less work than its headline rate suggests if retailers under-shift it, and the Ministry should monitor shelf prices before assuming the new base will raise as much as the old one per liter.

The natural next step for this project is to add quantities. Mexico's ENIGH household expenditure survey, which the published papers use [24], would let the price response estimated here

be turned into a demand elasticity and a welfare calculation. That is the boundary of the present result: this paper measures what happened to prices, and prices alone determine who pays.

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